

Gilbert Strang - A First Course in Calculus

Chapters 1 - 7

1 Chapter 1: Introduction to Vectors

1.1 Vectors and Linear Combinations

Problems 1-9 are about addition of vectors and linear combinations.

1. Describe geometrically (line, plane, or all of R^3) all linear combinations of

(a) $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$

(b) $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$

(c) $\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$

2. Draw $v = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$ and $v + w$ and $v - w$ in a single xy plane.

3. If $v + w = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$ and $v - w = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$, compute and draw the vectors v and w .

4. From $v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, find the components of $3v + w$ and $cv + dw$.

5. Compute $u + v + w$ and $2u + 2v + w$. How do you know u, v, w lie in a plane?

These lie in a plane because
 $w = cu + dv$. Find c and d

$$u = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad v = \begin{bmatrix} -3 \\ 1 \\ -2 \end{bmatrix}, \quad w = \begin{bmatrix} 2 \\ -3 \\ -1 \end{bmatrix}.$$

6. Every combination of $v = (1, -2, 1)$ and $w = (0, 1, -1)$ has components that add to _____.
Find c and d so that $cv + dw = (3, 3, -6)$. Why is $(3, 3, 6)$ impossible?

7. In the xy plane mark all nine of these linear combinations:

$$c \begin{bmatrix} 2 \\ 1 \end{bmatrix} + d \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{with} \quad c = 0, 1, 2 \quad \text{and} \quad d = 0, 1, 2.$$

8. The parallelogram in Figure 1.1 has diagonal $v + w$. What is its other diagonal? What is the sum of the two diagonals? Draw that vector sum.

9. If three corners of a parallelogram are $(1, 1)$, $(4, 2)$, and $(1, 3)$, what are all three of the possible fourth corners? Draw two of them.

Problems 10–14 are about special vectors on cubes and clocks in Figure 1.4.

10. Which point of the cube is $i + j$? Which point is the vector sum of $i = (1, 0, 0)$ and $j = (0, 1, 0)$ and $k = (0, 0, 1)$? Describe all points (x, y, z) in the cube.
11. Four corners of this unit cube are $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$. What are the other four corners? Find the coordinates of the center point of the cube. The center points of the six faces are _____. The cube has how many edges?
12. Review Question. In xyz space, where is the plane of all linear combinations of $i = (1, 0, 0)$ and $i + j = (1, 1, 0)$?

13. (a) What is the sum V of the twelve vectors that go from the center of a clock to the hours 1:00, 2:00, ..., 12:00?
- (b) If the 2:00 vector is removed, why do the 11 remaining vectors add to 8:00?
- (c) What are the x, y components of that 2:00 vector $v = (\cos \theta, \sin \theta)$?
14. Suppose the twelve vectors start from 6:00 at the bottom instead of $(0, 0)$ at the center. The vector to 12:00 is doubled to $(0, 2)$. The new twelve vectors add to _____.

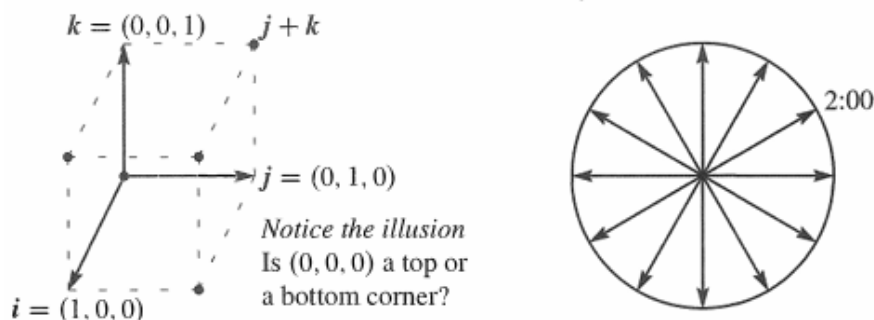


Figure 1.4: Unit cube from i, j, k and twelve clock vectors.

Problems 15–19 go further with linear combinations of v and w (Figure 1.5a).

15. Figure 1.5a shows $\frac{1}{2}v + \frac{1}{2}w$. Mark the points $\frac{3}{4}v + \frac{1}{4}w$ and $\frac{1}{4}v + \frac{3}{4}w$ and $v + w$.
16. Mark the point $-v + 2w$ and any other combination $cv + dw$ with $c + d = 1$. Draw the line of all combinations that have $c + d = 1$.
17. Locate $\frac{1}{3}v + \frac{1}{3}w$ and $\frac{2}{3}v + \frac{2}{3}w$. The combinations $cv + cw$ fill out what line?
18. Restricted by $0 \leq c \leq 1$ and $0 \leq d \leq 1$, shade in all combinations $cv + dw$.
19. Restricted only by $c \geq 0$ and $d \geq 0$ draw the “cone” of all combinations $cv + dw$.

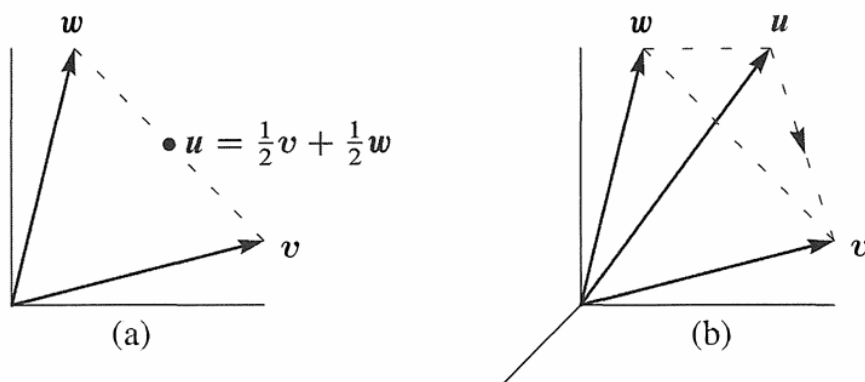


Figure 1.5: Problems 15-19 in a plane

Problems 20-25 in 3-dimensional space

Problems 20-25 deal with u, v, w in three-dimensional space (see Figure 1.5b).

20. Locate $\frac{1}{2}v + \frac{1}{2}w + \frac{1}{2}v$ and $\frac{1}{2}v + \frac{1}{2}w + \frac{1}{2}w$ in Figure 1.5b. Challenge problem: Under what restrictions on c, d, e , will the combinations $cu + dv + ew$ fill in the dashed triangle? To stay in the triangle, one requirement is $c \geq 0, d \geq 0, e \geq 0$.
21. The three sides of the dashed triangle are $v - u$ and $w - v$ and $u - w$. Their sum is _____. Draw the head-to-tail addition around a plane triangle of $(3, 1)$ plus $(-1, 1)$ plus $(-2, -2)$.

22. Shade in the pyramid of combinations $cu + dv + ew$ with $c \geq 0, d \geq 0, e \geq 0$ and $c + d + e \leq 1$. Mark the vector $\frac{1}{2}(u + v + w)$ as inside or outside this pyramid.
23. If you look at all combinations of those u, v , and w , is there any vector that can't be produced from $cu + dv + ew$? Different answer if u, v, w are all in _____.
24. Which vectors are combinations of u and v , and also combinations of v and w ?
25. Draw vectors u, v, w so that their combinations $cu + dv + ew$ fill only a line. Find vectors u, v, w so that their combinations $cu + dv + ew$ fill only a plane.
26. What combination $c \begin{bmatrix} 2 \\ 1 \end{bmatrix} + d \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ produces $\begin{bmatrix} 14 \\ 8 \end{bmatrix}$? Express this question as two equations for the coefficients c and d in the linear combination.

Challenge Problems

27. How many corners does a cube have in 4 dimensions? How many 3D faces? How many edges? A typical corner is $(0, 0, 1, 0)$. A typical edge goes to $(0, 1, 0, 0)$.
28. Find vectors v and w so that $v + w = (4, 5, 6)$ and $v - w = (2, 5, 8)$. This is a question with _____ unknown numbers, and an equal number of equations to find those numbers.
29. Find two different combinations of the three vectors $u = (1, 3)$ and $v = (2, 7)$ and $w = (1, 5)$ that produce $b = (0, 1)$. Slightly delicate question: If I take any three vectors u, v, w in the plane, will there always be two different combinations that produce $b = (0, 1)$?
30. The linear combinations of $v = (a, b)$ and $w = (c, d)$ fill the plane unless _____. Find four vectors u, v, w, z with four components each so that their combinations $cu + dv + ew + fz$ produce all vectors (b_1, b_2, b_3, b_4) in four-dimensional space.
31. Write down three equations for c, d, e so that $cu + dv + ew = b$. Can you somehow find c, d, e for b ?

$$u = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \quad v = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

1.2 Lengths and Dot Products

1. Calculate the dot products $u \cdot v$ and $u \cdot w$ and $u \cdot (v + w)$ and $w \cdot v$:

$$u = \begin{bmatrix} -6 \\ .8 \end{bmatrix}, \quad v = \begin{bmatrix} 4 \\ 3 \end{bmatrix}, \quad w = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

2. Compute the lengths $\|u\|$ and $\|v\|$ and $\|w\|$ of those vectors. Check the Schwarz inequalities $|u \cdot v| \leq \|u\| \|v\|$ and $v \cdot w \leq \|v\| \|w\|$.
3. Find unit vectors in the directions of v and w in Problem 1, and the cosine of the angle θ . Choose vectors a, b, c that make $0^\circ, 90^\circ$, and 180° angles with w .
4. For any *unit* vectors v and w , find the dot products (actual numbers) of
- (a) v and $-v$ (b) $v + w$ and $v - w$ (c) $v - 2w$ and $v + 2w$
5. Find unit vectors u_1 and u_2 in the directions of $v = (1, 3)$ and $w = (2, 1, 2)$. Find unit vectors U_1 and U_2 that are perpendicular to u_1 and u_2 .
6. (a) Describe every vector $w = (w_1, w_2)$ that is perpendicular to $v = (2, -1)$.
 (b) All vectors perpendicular to $V = (1, 1, 1)$ lie on a _____ in 3 dimensions.
 (c) The vectors perpendicular to both $(1, 1, 1)$ and $(1, 2, 3)$ lie on a _____.
7. Find the angle θ (from its cosine) between these pairs of vectors:

$$(a) \ v = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix} \text{ and } w = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$(b) \ v = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} \text{ and } w = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$$

$$(c) \ v = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix} \text{ and } w = \begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix}$$

$$(d) \ v = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \text{ and } w = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$$

8. True or false (give a reason if true or find a counterexample if false):

(a) If $u = (1, 1, 1)$ is perpendicular to v and w , then v is parallel to w .

(b) If u is perpendicular to v and w , then u is perpendicular to $v + 2w$.

(c) If u and v are perpendicular unit vectors then $\|u - v\| = \sqrt{2}$. Yes!

9. The slopes of the arrows from $(0, 0)$ to (v_1, v_2) and (w_1, w_2) are v_2/v_1 and w_2/w_1 . Suppose the product v_2w_2/v_1w_1 of those slopes is -1 . Show that $v \cdot w = 0$ and the vectors are perpendicular. (The line $y = 4x$ is perpendicular to $y = -\frac{1}{4}x$.)

10. Draw arrows from $(0, 0)$ to the points $v = (1, 2)$ and $w = (-2, 1)$. Multiply their slopes. That answer is a signal that $v \cdot w = 0$ and the arrows are _____.

11. If $v \cdot w$ is negative, what does this say about the angle between v and w ? Draw a 3-dimensional vector v (an arrow), and show where to find all w 's with $v \cdot w < 0$.

12. With $v = (1, 1)$ and $w = (1, 5)$ choose a number c so that $w - cv$ is perpendicular to v . Then find the formula for c starting from any nonzero v and w .

13. Find nonzero vectors v and w that are perpendicular to $(1, 0, 1)$ and to each other.

14. Find nonzero vectors u, v, w that are perpendicular to $(1, 1, 1)$ and to each other.

15. The geometric mean of $x = 2$ and $y = 8$ is $\sqrt{xy} = 4$. The arithmetic mean is larger: $\frac{1}{2}(x + y) = \frac{1}{2}(2 + 8) = 5$. This would come in Example 6 from the Schwarz inequality for $v = (\sqrt{2}, \sqrt{8})$ and $w = (\sqrt{8}, \sqrt{2})$. Find $\cos \theta$ for this v and w .

16. How long is the vector $v = (1, 1, \dots, 1)$ in 9 dimensions? Find a unit vector u in the same direction as v and a unit vector w that is perpendicular to v .

17. What are the cosines of the angles α, β, θ between the vector $(1, 0, -1)$ and the unit vectors i, j, k along the axes? Check the formula $\cos^2 \alpha + \cos^2 \beta + \cos^2 \theta = 1$.

Problems 18-28 lead to the main facts about lengths and angles in triangles.

18. The parallelogram with sides $v = (4, 2)$ and $w = (-1, 2)$ is a rectangle. Check the Pythagoras formula $a^2 + b^2 = c^2$ which is for *right triangles only*:

$$(\text{length of } v)^2 + (\text{length of } w)^2 = (\text{length of } v + w)^2.$$

19. (Rules for dot products) These equations are simple but useful:

$$(a) \ v \cdot w = w \cdot v$$

$$(b) \ u \cdot (v + w) = u \cdot v + u \cdot w$$

$$(c) \ (cv) \cdot w = c(v \cdot w)$$

Use (2) with $u = v + w$ to prove $\|v + w\|^2 = v \cdot v + 2v \cdot w + w \cdot w$.

20. The "Law of Cosines" comes from $(v - w) \cdot (v - w) = v \cdot v - 2v \cdot w + w \cdot w$:

$$\text{Cosine Law} \quad \|v - w\|^2 = \|v\|^2 - 2\|v\|\|w\|\cos \theta + \|w\|^2.$$

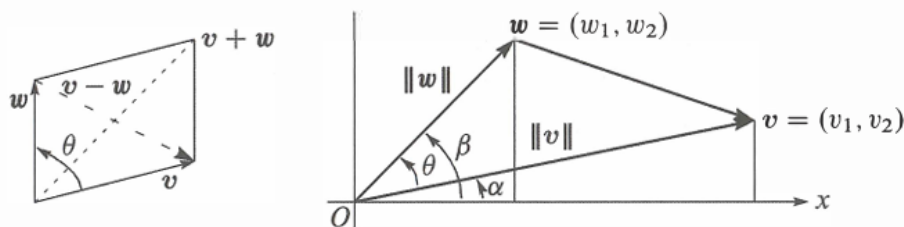
Draw a triangle with sides v and w and $v - w$. Which of the angles is θ ?

21. The triangle inequality says: (length of $v + w$) $\leq$ (length of v) + (length of w).

Problem 19 found $\|v + w\|^2 = \|v\|^2 + 2v \cdot w + \|w\|^2$.

Increase by $\|v\| \|w\|$ to show that $v \cdot w$ can not exceed $\|v\| \|w\|$:

Triangle inequality $\|v + w\|^2 \leq (\|v\| + \|w\|)^2$ or $\boxed{\|v + w\| \leq \|v\| + \|w\|}$.



22. The Schwarz inequality $|v \cdot w| \leq \|v\| \|w\|$ by algebra instead of trigonometry:

(a) Multiply out both sides of $(v_1 w_1 + v_2 w_2)^2 \leq (v_1^2 + v_2^2)(w_1^2 + w_2^2)$.

(b) Show that the difference between those two sides equals $(v_1 w_2 - v_2 w_1)^2$. This cannot be negative since it is a square—so the inequality is true.

23. The figure shows that $\cos \alpha = v_1 / \|v\|$ and $\sin \alpha = v_2 / \|v\|$. Similarly $\cos \beta$ is _____ and $\sin \beta$ is _____. The angle θ is $\beta - \alpha$. Substitute into the trigonometry formula $\cos \beta \cos \alpha + \sin \beta \sin \alpha$ for $\cos(\beta - \alpha)$ to find $\cos \theta = v \cdot w / \|v\| \|w\|$.

24. One-line proof of the inequality $|u \cdot U| \leq 1$ for unit vectors (u_1, u_2) and (U_1, U_2) :

$$|u \cdot U| \leq |u_1| |U_1| + |u_2| |U_2| \leq \frac{u_1^2 + U_1^2}{2} + \frac{u_2^2 + U_2^2}{2} = 1.$$

Put $(u_1, u_2) = (.6, .8)$ and $(U_1, U_2) = (.8, .6)$ in that whole line and find $\cos \theta$.

25. Why is $|\cos \theta|$ never greater than 1 in the first place?

26. (Recommended) Draw a parallelogram.

27. Parallelogram with two sides v and w . Show that the squared diagonal lengths $\|v + w\|^2 + \|v - w\|^2$ add to the sum of four squared side lengths $2\|v\|^2 + 2\|w\|^2$.

28. If $v = (1, 2)$ draw all vectors $w = (x, y)$ in the xy -plane with $v \cdot w = x + 2y = 5$. Why do those w 's lie along a line? Which is the shortest w ?

29. (Recommended) If $\|v\| = 5$ and $\|w\| = 3$, what are the smallest and largest possible values of $\|v - w\|$? What are the smallest and largest possible values of $v \cdot w$?

Challenge Problems

30. Can three vectors in the xy -plane have $u \cdot v < 0$ and $v \cdot w < 0$ and $u \cdot w < 0$?

I don't know how many vectors in xyz space can have all negative dot products. (Four of those vectors in the plane would certainly be impossible....)

31. Pick any numbers that add to $x + y + z = 0$. Find the angle between your vector $v = (x, y, z)$ and the vector $w = (z, x, y)$. Challenge question: Explain why $v \cdot w / \|v\| \|w\|$ is always $-\frac{1}{2}$.

32. How could you prove $\sqrt[3]{xyz} \leq \frac{1}{3}(x + y + z)$ (geometric mean $\leq$ arithmetic mean)?

33. Find 4 perpendicular unit vectors of the form $(\pm \frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2})$: Choose + or -.

34. Using $v = \text{randn}(3, 1)$ in MATLAB, create a random unit vector $u = v / \|v\|$.

Using $V = \text{randn}(3, 30)$ create 30 more random unit vectors U_j .

What is the average size of the dot products $|u \cdot U_j|$?

In calculus, the average is $\int_0^\pi |\cos \theta| d\theta / \pi = 2/\pi$.

1.3 Matrices

1. Find the linear combination $3s_1 + 4s_2 + 5s_3 = b$. Then write b as a matrix-vector multiplication Sx , with 3, 4, 5 in x . Compute the three dot products (row of S) $\cdot x$:

$$s_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad s_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad s_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \text{go into the columns of } S.$$

2. Solve these equations $Sy = b$ with s_1, s_2, s_3 in the columns of S :

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 9 \end{bmatrix}.$$

S is a sum matrix. The sum of the first 3 odd numbers is _____.

3. Solve these three equations for y_1, y_2, y_3 in terms of c_1, c_2, c_3 :

$$Sy = c \quad \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

Write the solution y as a matrix $A = S^{-1}$ times the vector c . Are the columns of S independent or dependent?

4. Find a combination $x_1w_1 + x_2w_2 + x_3w_3$ that gives the zero vector with $x_1 = 1$:

$$w_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad w_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \quad w_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$$

Those vectors are (independent) (dependent). The three vectors lie in a _____. The matrix W with those three columns is not invertible.

5. The rows of that matrix W produce three vectors (I write them as columns):

$$r_1 = \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} \quad r_2 = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} \quad r_3 = \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$$

Linear algebra says that these vectors must also lie in a plane. There must be many combinations with $y_1r_1 + y_2r_2 + y_3r_3 = 0$. Find two sets of y 's.

6. Which numbers c give dependent columns so a combination of columns equals zero?

$$\begin{bmatrix} 1 & 1 & 0 \\ 3 & 2 & 1 \\ 7 & 4 & c \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & c \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} c & c & c \\ 2 & 1 & 5 \\ 3 & 3 & 6 \end{bmatrix} \quad \begin{array}{l} \text{maybe} \\ \text{always} \\ \text{independent for } c \neq 0 \end{array}$$

7. If the columns combine into $Ax = 0$, then each of the rows has $r \cdot x = 0$:

$$\begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{By rows} \quad \begin{bmatrix} r_1 \cdot x \\ r_2 \cdot x \\ r_3 \cdot x \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The three rows also lie in a plane. Why is that plane perpendicular to x ?

8. Moving to a 4×4 difference equation $Ax = b$, find the four components x_1, x_2, x_3, x_4 . Then write this solution as $x = A^{-1}b$ to find the inverse matrix:

$$Ax = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = b.$$

9. What is the cyclic 4 by 4 difference matrix C ? It will have 1 and -1 in each row and each column. Find all solutions $x = (x_1, x_2, x_3, x_4)$ to $Cx = 0$. The four columns of C lie in a “three-dimensional hyperplane” inside four-dimensional space.

10. A forward difference matrix Δ is upper triangular:

$$\Delta z = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} z_2 - z_1 \\ z_3 - z_2 \\ 0 - z_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = b.$$

Find z_1, z_2, z_3 from b_1, b_2, b_3 . What is the inverse matrix in $z = \Delta^{-1}b$?

11. Show that the forward differences $(t+1)^2 - t^2$ are $2t+1 = \text{odd numbers}$. As in calculus, the difference $(t+1)^n - t^n$ will begin with the derivative of t^n , which is _____.

12. The last lines of the Worked Example say that the 4×4 centered difference matrix in (16) is invertible. Solve $Cx = (b_1, b_2, b_3, b_4)$ to find its inverse in $x = C^{-1}b$.

Challenge Problems

13. The very last words say that the 5×5 centered difference matrix is not invertible. Write down the 5 equations $Cx = b$. Find a combination of left sides that gives zero. What combination of b_1, b_2, b_3, b_4, b_5 must be zero? (The 5 columns lie on a “4-dimensional hyperplane” in 5-dimensional space. Hard to visualize.)

14. If (a, b) is a multiple of (c, d) with $abcd \neq 0$, show that (a, c) is a multiple of (b, d) . This is surprisingly important: two columns are falling on one line. You could use numbers first to see how a, b, c, d are related. The question will lead to:

If $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has dependent rows, then it also has dependent columns.

2 Chapter 2: Solving Linear Equations

2.1 Vectors and Linear Equations

Challenge Problems

2.2 The Idea of Elimination

Challenge Problems

2.3 Elimination Using Matrices

Challenge Problems

2.4 Rules for Matrix Operations

Challenge Problems

2.5 Inverse Matrices

Challenge Problems

2.6 Elimination = Factorization: $A = LU$

Challenge Problems

2.7 Transposes and Permutations

Challenge Problems

3 Chapter 3: Vector Spaces and Subspaces

3.1 Spaces of Vectors

Challenge Problems

3.2 The Nullspace of A : Solving $Ax = 0$ and $Rx = 0$

Challenge Problems

3.3 The Complete Solution to $Ax = b$

Challenge Problems

3.4 Independence, Basis and Dimension

Challenge Problems

3.5 Dimensions of the Four Subspaces

Challenge Problems

4 Chapter 4: Orthogonality

4.1 Orthogonality of the Four Subspaces

Challenge Problems

4.2 Projections

Challenge Problems

4.3 Least Squares Approximations

Challenge Problems

4.4 Orthonormal Bases and Gram-Schmidt

Challenge Problems

5 Chapter 5: Determinants

5.1 The Properties of Determinants

Challenge Problems

5.2 Permutations and Cofactors

Challenge Problems

5.3 Cramer's Rule, Inverses, and Volumes

Challenge Problems

6 Chapter 6: Eigenvalues and Eigenvectors

6.1 Introduction to Eigenvalues

Challenge Problems

6.2 Diagonalizing a Matrix

Challenge Problems

6.3 Systems of Differential Equations

Challenge Problems

6.4 Symmetric Matrices

Challenge Problems

6.5 Positive Definite Matrices

Challenge Problems

7 Chapter 7: The Singular Value Decomposition (SVD)

7.1 Image Processing by Linear Algebra

Challenge Problems

7.2 Bases and Matrices in the SVD

Challenge Problems

7.3 Principal Component Analysis (PCA by the SVD)

Challenge Problems

7.4 The Geometry of the SVD

Challenge Problems